Outliers, Leverage, and Influence

Applied Linear Models I • Lecture 7 of 16

Applied Linear Models I

Three different questions

- **Leverage**: is the point unusual in X-space?
- **Outlier**: is y far from the prediction?
- **Influence**: does deleting the point change the fit?

Central message

A point moves the fit only when **high leverage and outlyingness coincide**.

Leverage: position in X-space

$h_{ii} = [\mathrm{H}]_{ii} = \partial \hat{y}_i / \partial y_i$. Determined by x_i alone.

$$\tfrac{1}{n} \le h_{ii} \le 1, \qquad \sum_i h_{ii} = \mathrm{tr}(\mathrm{H}) = p, \qquad \text{flag } h_{ii} > \frac{2p}{n}.$$

- Average leverage is p/n.
- `LifeCycleSavings` (p=5, n=50, cutoff 0.20): Libya, United States are highest-leverage.
- High leverage is *potential* influence – not yet realized.

Outliers: extreme in y

Raw residuals are heteroskedastic; use the **studentized deleted residual**

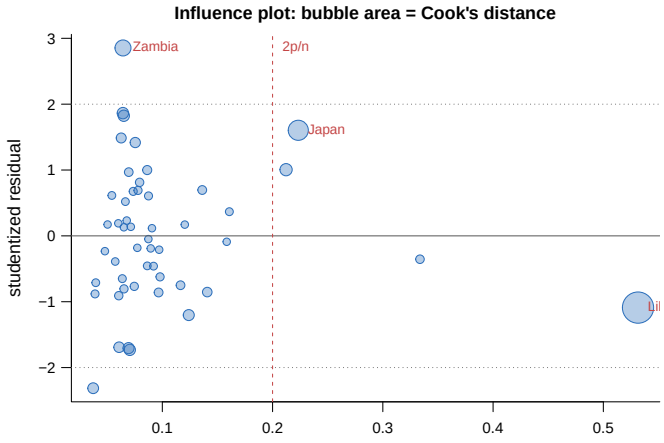
$$t_i = \frac{e_i}{\hat{\sigma}_{(i)}\sqrt{1 - h_{ii}}} \sim t_{n-p-1},$$

so a true outlier cannot mask itself. Screen all n points with a **Bonferroni** cutoff $t_{n-p-1,\,1-\alpha/(2n)} \approx 3.5$.

Savings data: Zambia is the most extreme ($t \approx 2.85$) – notable, below the bar.

Influence: leverage × outlyingness

$$D_i = \frac{r_i^2}{p} \cdot \frac{h_{ii}}{1 - h_{ii}} \qquad \text{(Cook's distance, flag } D_i > 4/n\text{)}.$$

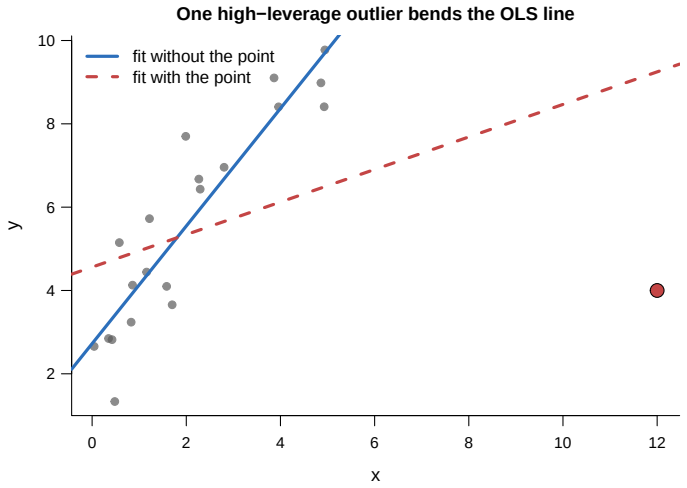


The three rankings differ

Rank	Leverage	Outlyingness	Influence
1	Libya	Zambia	Libya
2	United States	Chile	Japan
3	Japan	Philippines	Zambia

- **Zambia**: biggest outlier, only 3rd in influence (ordinary leverage).
- **United States**: high leverage, not top-3 influence (small residual).
- **Libya**: tops both $\Rightarrow$ the influential point.

One point can bend the line



A high-leverage *outlier* swings the slope; the same residual at average leverage would not.

What to do – responsibly

Refit without Libya: the `ddpi` coefficient changes by $\approx 49\%$.

1. **Investigate the case** – error? different population? rare-but-real?
2. **Report sensitivity** – show results with and without.
3. **Delete only with justification**; or down-weight via robust regression (Lecture 15).

Never

Silently drop inconvenient points – it is data manipulation and fakes precision.

Summary

- Leverage = X-position ($> 2p/n$); outliers = big studentized residuals.
- Influence = leverage $\times$ outlyingness (Cook's $D > 4/n$).
- The three are distinct – only their coincidence moves the fit.
- Investigate and report; do not delete silently.

Next

Lecture 8: standardized and no-intercept models (then the midterm).